

Developer Mock Test Results

Test ID:	43b9571e-239e-41...	Student ID:	81840
Test Type:	Developer Assessment	Date:	2026-03-04 11:07:27
Overall Score:	6/25 (24.0%)	Performance:	Needs Improvement
Warnings:	0/3	Status:	Completed

Section-wise Performance

Section	Score	Percentage	Status
Aptitude	0/10	0.0%	Needs Work
MCQ/Theory	6/10	60.0%	Pass
Coding	0/5	0.0%	Needs Work

Detailed Question Review

APTITUDE SECTION (0/10 - 0.0%)

Q1. If a product is discounted by 10%, and then further discounted by 20%, what is the total percentage discount?

Your Answer: SKIPPED

Correct Answer: 28%

To find the total percentage discount, first, calculate the price after the initial 10% discount: if the original price is \$100, the price after the 10% discount is $\$100 - (10\% \text{ of } \$100) = \$100 - \$10 = \$90$. Then, apply the 20% discount on the \$90 price: 20% of \$90 is \$18, so the final price is $\$90 - \$18 = \$72$. The total discount is $\$100 - \$72 = \$28$, which is 28% of the original price. The key calculation step is: $\$100 - (\$100 * 0.10) = \$90$, then $\$90 - (\$90 * 0.20) = \$72$, and $(\$100 - \$72) / \$100 = 0.28$ or 28%. This explanation assumes the user may have made a mistake by not calculating the second discount based on the already discounted price, instead

Q2. The average of 8 numbers is 24. If the sum of 3 of these numbers is 60, what is the average of the remaining 5 numbers?

Your Answer: SKIPPED

Correct Answer: 22

To find the average of the remaining 5 numbers, we first calculate the total sum of the 8 numbers, which is $8 * 24 = 192$. Since the sum of 3 of these numbers is 60, the sum of the remaining 5 numbers is $192 - 60 = 132$. The average of the remaining 5 numbers is then $132 / 5 = 26.4$, but since this option is not available, we re-evaluate the given options and calculations. Given the provided options and the calculation, it seems there might have been a miscalculation in the interpretation of the provided correct answer. The correct step is calculating the total sum and then subtracting the sum of the 3 numbers to find the sum of the remaining 5, which we did. However, the provided correct answer is 22, which would imply a different calculation or interpretation of the question's data. The likely mistake, if following the provided correct

Q3. A shopkeeper sells a product at Rs.100, which was bought for Rs.80. What is the profit percentage?

Your Answer: __SKIPPED__

Correct Answer: 20%

■ To find the profit percentage, we calculate the profit made (selling price - cost price) and divide it by the cost price, then multiply by 100. The key calculation step is: $(100 - 80) / 80 * 100 = 20 / 80 * 100 = 20\%$. The correct answer is 20% because the profit made (Rs.20) is 20% of the cost price (Rs.80), indicating the user likely didn't perform this calculation if they got it wrong.

■ **Q4. A seller sells a product at a 15% profit. If the cost price is Rs.600, what is the selling price?**

Your Answer: __SKIPPED__

Correct Answer: Rs.690

■ To find the selling price, we need to calculate the profit first, which is 15% of the cost price (Rs.600). The key calculation step is: Profit = 15% of Rs.600 = $0.15 * 600 = \text{Rs.}90$. Then, the selling price is the cost price plus the profit: Selling Price = Rs.600 + Rs.90 = Rs.690. (Note: The user's answer was skipped, so we can't comment on their likely mistake.)

■ **Q5. A seller sells a product at a 20% loss. If the selling price is Rs.800, what is the cost price?**

Your Answer: __SKIPPED__

Correct Answer: Rs.1000

■ To find the cost price, we can use the formula: Cost Price = Selling Price / (1 - Loss Percentage). The key calculation step is: Cost Price = $800 / (1 - 0.20) = 800 / 0.80 = 1000$. Therefore, the correct answer is Rs.1000, as the seller sold the product at a 20% loss, resulting in a cost price of Rs.1000. (Note: The user's answer was skipped, so we can't determine their likely mistake.)

■ **Q6. If 5 men can do a piece of work in 10 days, how many men would be required to do the same work in 5 days?**

Your Answer: __SKIPPED__

Correct Answer: 10

■ To solve this problem, we can use the concept of work and time inversely related to the number of men. The key calculation step is: if 5 men can do the work in 10 days, then the total work done is 5 men * 10 days = 50 man-days. To do the same work in 5 days, we need to find the number of men required, which is 50 man-days / 5 days = 10 men. The correct answer is indeed 10, as calculated above. The user likely made a mistake by not considering the inverse relationship between the number of men and the time required to complete the work.

■ **Q7. If all A are B, and all B are C, what can be concluded?**

Your Answer: __SKIPPED__

Correct Answer: All A are C

■ The correct answer is "All A are C" because if all A are B and all B are C, then it logically follows that all A are also C. This is a classic example of a syllogism, where the transitive property of inclusion is applied. The user's likely mistake would be misunderstanding the transitive property, which allows us to conclude that A is a subset of C, given that A is a subset of B and B is a subset of C.

■ **Q8. What is the next number: 3, 6, 12, 24, 48, ?**

Your Answer: __SKIPPED__

Correct Answer: 96

■ The correct answer is 96 because the pattern of the sequence is obtained by multiplying the previous term by 2. The key calculation step is: $48 * 2 = 96$. This indicates that the sequence is formed by doubling the previous term, making 96 the next number in the sequence.

■ **Q9. What is the next number: 2, 6, 12, 20, 30, ?**

Your Answer: __SKIPPED__

Correct Answer: 42

■ The correct answer is 42 because the pattern of the sequence is obtained by adding 4, 6, 8, 10, ... to the previous term. The key calculation step is: $2 + 4 = 6$, $6 + 6 = 12$, $12 + 8 = 20$, $20 + 10 = 30$, $30 + 12 = 42$. This sequence is formed by adding consecutive even numbers, which is why 42 is the next number in the sequence.

■ **Q10. A bag contains Rs.1, Rs.2, and Rs.5 coins in the ratio 3:4:5. If the total amount is Rs.120, how many coins of each type are there?**

Your Answer: __SKIPPED__

Correct Answer: 15, 20, 25

■ Let's denote the number of Rs.1, Rs.2, and Rs.5 coins as $3x$, $4x$, and $5x$, respectively. The total amount is Rs.120, so the equation becomes: $1(3x) + 2(4x) + 5(5x) = 120$, which simplifies to $3x + 8x + 25x = 120$, then $36x = 120$, and $x = 120 / 36 = 10/3$. However, since the correct answer is given as 15, 20, 25, let's re-evaluate: if we multiply the ratio by 5 (to get whole numbers), we have 15, 20, 25, and the total amount equation becomes: $1(15) + 2(20) + 5(25) = 15 + 40 + 125 = 180$, which

■ MCQ SECTION (6/10 - 60.0%)

■ Q1. What is Java primarily used for?

Your Answer: Web development, mobile apps, and enterprise software

■ Well done!

■ Q2. What is a package in Java?

Your Answer: A collection of related classes

■ Well done!

■ Q3. What is the file extension for a Java file?

Your Answer: .java

■ Correct! Well done.

■ Q4. What is the purpose of a loop in Java?

Your Answer: To repeat a set of instructions

■ Excellent! Right answer.

■ Q5. What is the purpose of compiling a Java program?

Your Answer: To run the program

Correct Answer: To convert the code into an executable file

■ The correct answer, "To convert the code into an executable file", is right because compiling a Java program translates the Java source code into an intermediate format called bytecode, which can be executed by the Java Virtual Machine (JVM). This step is necessary to prepare the code for execution. The user's answer, "To run the program", is incorrect because while compilation is a prerequisite for running a Java program, its primary purpose is to convert the code, not to execute it directly.

■ Q6. What is the first step in writing a Java program?

Your Answer: Creating a new Java file

Correct Answer: Installing Java JDK

■ The correct answer is Installing Java JDK because, to write a Java program, you first need to have the Java Development Kit (JDK) installed on your computer, which provides the necessary tools and libraries to compile and run Java code. The user's answer, Creating a new Java file, is a step that comes later in the process, after the JDK is installed. The user likely mistook a subsequent step for the initial one, overlooking the need to set up the development environment before starting to code.

■ Q7. What is the purpose of a try-catch block in Java?

Your Answer:

To handle errors and exceptions

■ Correct! Well done.

■ Q8. Why is it important to test Java programs regularly?

Your Answer: To make the code more efficient

Correct Answer: To ensure the program works correctly

■ The correct answer, "To ensure the program works correctly," is right because regular testing helps identify and fix bugs, ensuring the program functions as intended and meets its requirements. This is crucial in software development to prevent errors and provide a reliable user experience. The user's answer, "To make the code more efficient," is incorrect because while testing can sometimes lead to efficiency improvements, its primary purpose is to verify the program's correctness, not optimize its performance.

■ Q9. What is the purpose of Java programming?

Your Answer: To build applications efficiently

■ Correct! Well done.

■ Q10. What are the types of data types in Java?

Your Answer: int, double, char, and String

Correct Answer: int, double, char, String, and arrays

■ The correct answer is B) int, double, char, String, and arrays because Java has two main categories of data types: primitive types (such as int, double, and char) and reference types (such as String and arrays). The user's answer is incorrect because it misses arrays, which are a fundamental data type in Java. The user likely overlooked the fact that arrays are a built-in data type in Java, not just a collection of other data types.

■ CODING SECTION (0/5 - 0.0%)

■ Q1. Write a Java program that reads a list of integers from stdin and filters out the even numbers. ****Input Format:**** - The first line: the number of int...

Your Answer:

__SKIPPED__

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        scanner.nextLine(); // Consume newline left-over

        StringBuilder output = new StringBuilder();
        for (int i = 0; i < n; i++) {
            int num = scanner.nextInt();
            scanner.nextLine(); // Consume newline left-over
            if (num % 2 != 0) {
                output.append(num).append(" ");
            }
        }

        System.out.println(output.toString().trim());
        scanner.close();
    }
}
```

■ There is no test case failure information provided to analyze and give feedback on. Please provide the test case results to determine the cause of the issue. Once the results are available, I can provide a brief explanation of what went wrong.

■ Q2. Write a Java program that reads an array of integers from stdin and calculates their sum. ****Input Format:**** - The first line: the number of integers ...

Your Answer:

__SKIPPED__

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        int sum = 0;
        for (int i = 0; i < n; i++) {
            sum += scanner.nextInt();
        }
        System.out.println(sum);
        scanner.close();
    }
}
```

■ There are no test case results provided to analyze the issue with the student's code. No feedback can be given without the actual test case results. Please provide the test case results to determine the cause of the failure.

■ **Q3. Write a Java program that simulates a bank account. It should read commands from stdin and perform the corresponding actions. **Input Format:** - The...**

Your Answer:

__SKIPPED__

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        double balance = scanner.nextDouble();
        scanner.nextLine(); // Consume newline left-over

        while (scanner.hasNextLine()) {
            String[] command = scanner.nextLine().split(" ");
            switch (command[0]) {
                case "deposit":
                    balance += Double.parseDouble(command[1]);
                    System.out.println("New balance: " + balance);
                    break;
                case "withdraw":
                    double amount = Double.parseDouble(command[1]);
                    if (amount > balance) {
                        System.out.println("Insufficient funds");
                    } else {
                        balance -= amount;
                        System.out.println("New balance: " + balance);
                    }
                    break;
                case "balance":
                    System.out.println("Current balance: " + balance);
                    break;
                default:
                    System.out.println("Invalid command");
            }
        }
        scanner.close();
    }
}
```

■ There are no test case results provided to analyze the issue with the student's code. No feedback can be given without the actual test case results. Please provide the test case results to determine the cause of the failure.

■ Q4. Write a Java program that reads a string and an integer from stdin, applies a Caesar cipher with the given shift, and prints the encrypted string. **...

Your Answer:

__SKIPPED__

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine();
        int shift = scanner.nextInt();
        scanner.close();

        StringBuilder encrypted = new StringBuilder();
        for (char c : input.toCharArray()) {
            if (Character.isLetter(c)) {
                char base = Character.isUpperCase(c) ? 'A' : 'a';
                encrypted.append((char) ((c - base + shift) % 26 + base));
            } else {
                encrypted.append(c);
            }
        }

        System.out.println(encrypted.toString());
    }
}
```

■ There are no test case results provided to analyze the issue with the student's code. No feedback can be given without the actual test case results. Please provide the test case results to determine the cause of the failure.

■ Q5. Write a Java program that reads a list of integers from stdin, sorts them in ascending order, and prints the sorted list. **Input Format:** - The fir...

Your Answer:

__FINAL_SUBMIT__

Correct Answer:

```
import java.util.Arrays;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        int[] list = new int[n];
        for (int i = 0; i < n; i++) {
            list[i] = scanner.nextInt();
        }
        Arrays.sort(list);
        for (int i = 0; i < n; i++) {
            System.out.print(list[i] + " ");
        }
        scanner.close();
    }
}
```

■ There is no test case failure information provided to analyze the issue with the student's code. No feedback can be given without the actual test case results. Please provide the test case results to determine the cause of the failure.

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.